

RiC Weekly Note #1

Lecture Content

- Introduction to the course
- Introduction to ROS

Literature

- ROS2 Introduction material:
<https://docs.ros.org/en/jazzy/Concepts/Basic.html>
We will focus on the topics until (including) *Services*.

Exercises

The best way of getting familiar with ROS is to explore it actively. The following is a suggestion for getting started:

1. Install a ROS2 version of your choice – we recommend using Ubuntu 24.04 and ROS2 Jazzy Jalisco. You can consult <https://docs.ros.org/en/jazzy/Installation.html> to see which version is recommended for which OS version. We will provide support for ROS2 on a native Ubuntu system (24.04). In case your operating system is not supported. There is a separate guide for the installation.
2. Explore the tutorials <https://docs.ros.org/en/jazzy/Tutorials.html> , in particular the following:
 - 2.1. Beginner, CLI Tools: Configuring environment
Use Domain ID 101 unless you prefer something else for a specific reason
 - 2.2. Beginner, CLI Tools: Turtlesim
 - 2.3. Beginner, CLI Tools: Understanding nodes / topics / services
 - 2.4. Beginner, Client Libraries: Using colcon, creating a workspace ...
 - 2.5. Beginner, Client Libraries: Writing your own publisher and subscriber
 - 2.6. Explore additional tutorials based on own interest
3. Write your own launch-file, e.g., launching above nodes, consider to
 - 3.1. Remap the topic
 - 3.2. Rename the nodes
4. Access a webcam (if you have one at hand) and inspect the image feed using RVIZ
This package is one option for accessing a webcam:
hint: The package `cam2image` can help access webcams
RVIZ can visualize *images* by subscribing to the correct topic
5. Run the example with the chatter-nodes on different computers, e.g., together with a coursemate
 - 5.1. Remember that the Domain ID needs to match.
You can read more about the Domain ID here:
<https://docs.ros.org/en/jazzy/Concepts/Intermediate/About-Domain-ID.html>

The above is considered a starting point for exploring ROS2. Please feel free to move on from here.

Programming Exercises 1

Will be provided in a separate document at the lecture.